

Time Series Clustering

Machine Learning for Behavioral Data

April 14, 2026

Today's Topic

Week	Lecture/Lab
9	Time Series Clustering
10	Explainability
11	Fairness
12	Reserve
13	No Lecture
14	No Lecture
15	Poster Session

Getting ready for today's lecture...

➤ Join us on SpeakUp

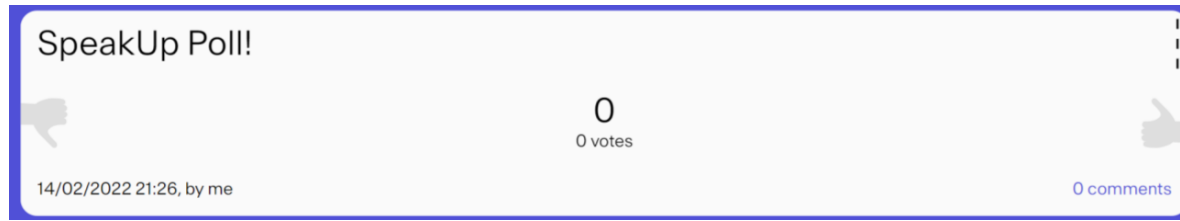
<https://go.epfl.ch/mlbd-speakup>



Short quiz about the past...

In K-Means clustering, which of the following parameters affect the goodness of the solution?

- a) Number of iterations
- b) Initial positioning of cluster centers
- c) Choice of k

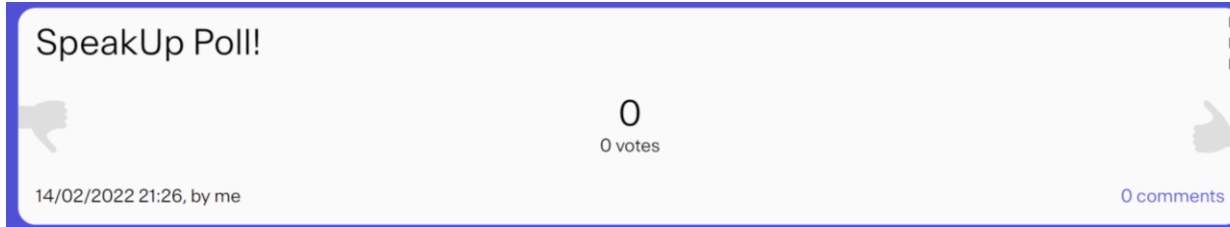


<https://go.epfl.ch/mlbd-speakup>

Short quiz about the past...

K-Means clustering is useful when dealing with non-convex clusters:

- a) True
- b) False

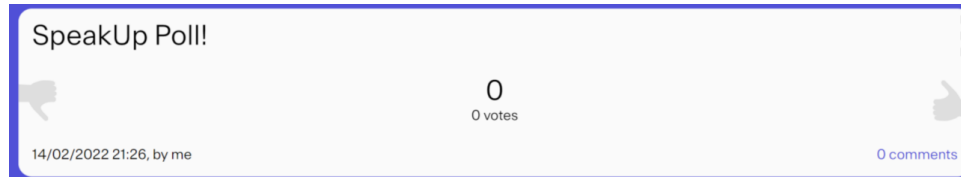


<https://go.epfl.ch/mlbd-speakup>

Short quiz about the past...

In a binary classification problem, it is appropriate to use the following activation function for the output layer:

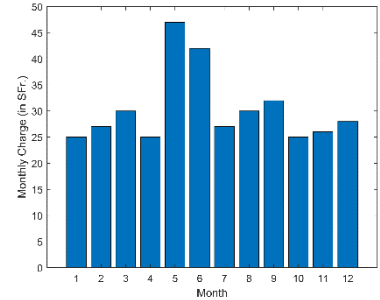
- a) Linear
- b) Tanh
- c) Sigmoid



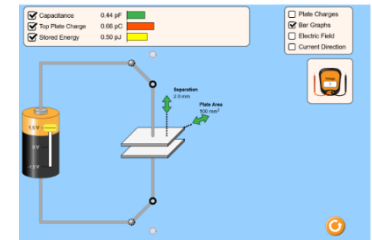
<https://go.epfl.ch/mlbd-speakup>

Today – Clustering Time Series Data

1. Aggregating features over time
2. Defining fixed time intervals (weeks, levels in a game, etc.)
3. Dynamic Time Warping



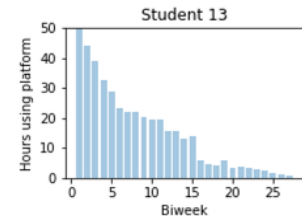
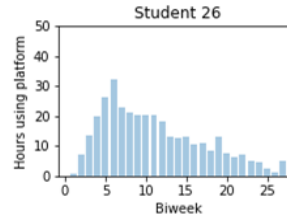
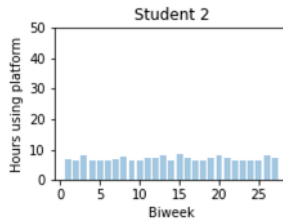
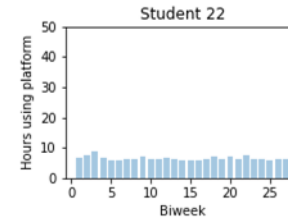
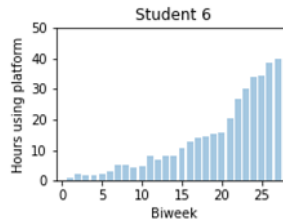
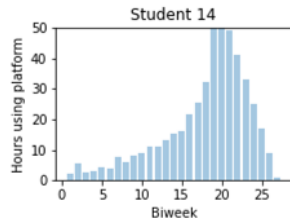
-
4. String Metrics
 5. Markov Models



Action Sequences

Today's Use Case

- Synthetic data of 30 high school students
- Time spent on an e-learning platform over one year (computed per biweek)
- Three clusters: 1) precrastinators, 2) regular, 3) procrastinators



Agenda

- **Aggregating features over time**
- Defining fixed time intervals (weeks, levels in a game, etc.)
- Dynamic Time Warping
- String Metrics
- Markov Models

Aggregating features over time

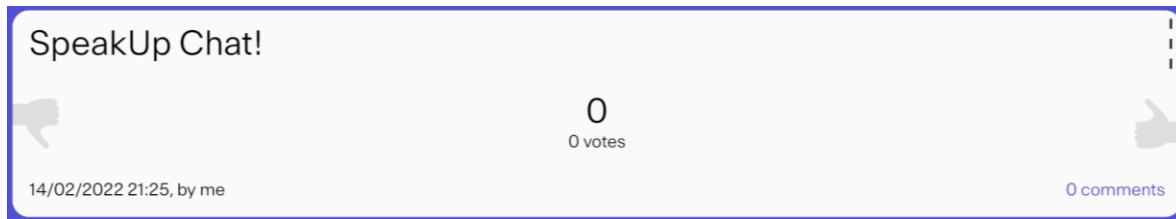
- We compute the value of the feature over the whole time series (average, maximum, range, standard deviation)
- We do not explicitly represent changes in features over time

➡ We can use standard distance/similarity measures

Your Turn – Aggregated Data

Run spectral clustering on the average number of hours:

- Can we interpret the different clusters?
- Are we able to retrieve the procrastination patterns? If not, why not?



<https://go.epfl.ch/mlbd-speakup>

Agenda

- Aggregating features over time
- **Defining fixed time intervals (weeks, levels in a game, etc.)**
- Dynamic Time Warping
- String Metrics
- Markov Models
- Additional Practice

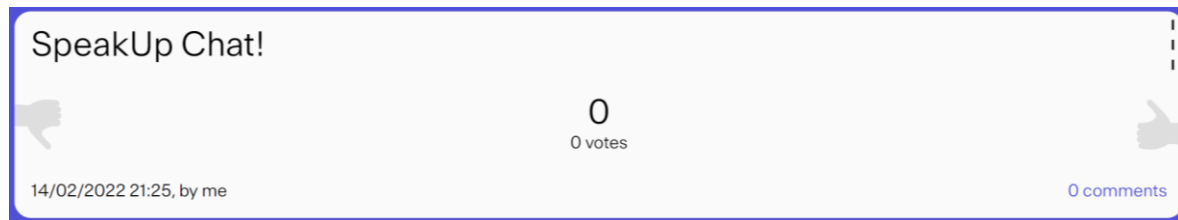
Using fixed time intervals

- Compute the feature value at fixed points in time (e.g., weeks, level in a game)
 - We obtain feature vectors with the same length for every student
- ➡ We can use standard distance measures

Your Turn – Fixed Time Intervals

Run spectral clustering on the vectors of biweeks (dimension = 27) using Euclidean distance:

- What is the optimal number of clusters?
- How do the results differ from the aggregated feature results?



<https://go.epfl.ch/mlbd-speakup>

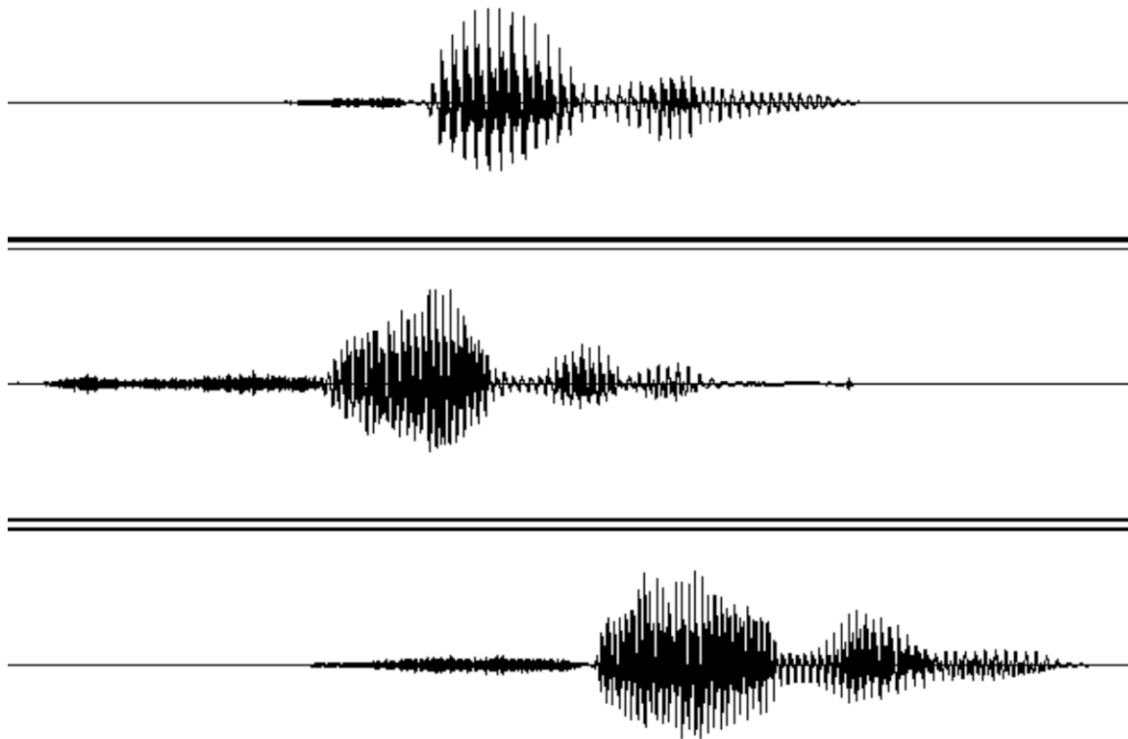
Agenda

- Aggregating features over time
- Defining fixed time intervals (weeks, levels in a game, etc.)
- **Dynamic Time Warping**
- String Metrics
- Markov Models
- Additional Practice (if time permits)

Dynamic Time Warping

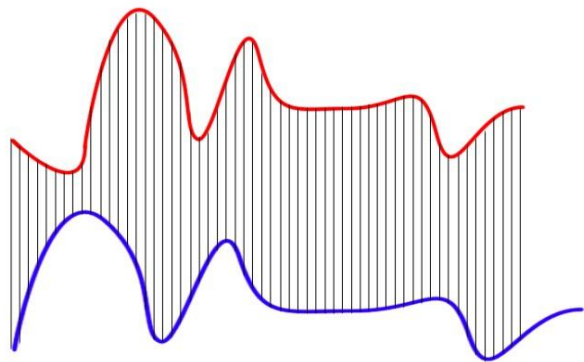
- Compute distance between two time series, which may vary in speed
- Time series can have different lengths
- ➡ Develop a **one-to-many** match, i.e. find an *optimal alignment* between two time series

Example: Spoken Digits

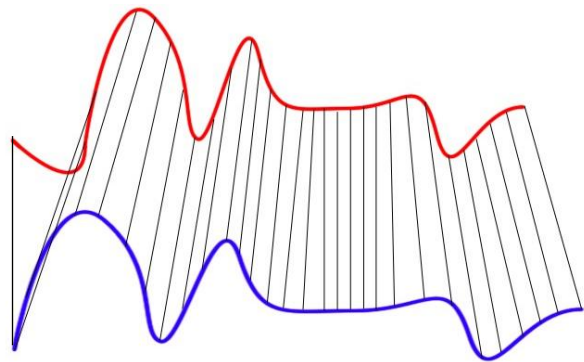


[Image Credit: CS 498, Signals AI, University of Illinois]

Dynamic Time Warping vs. Euclidean Distance



Euclidean Distance



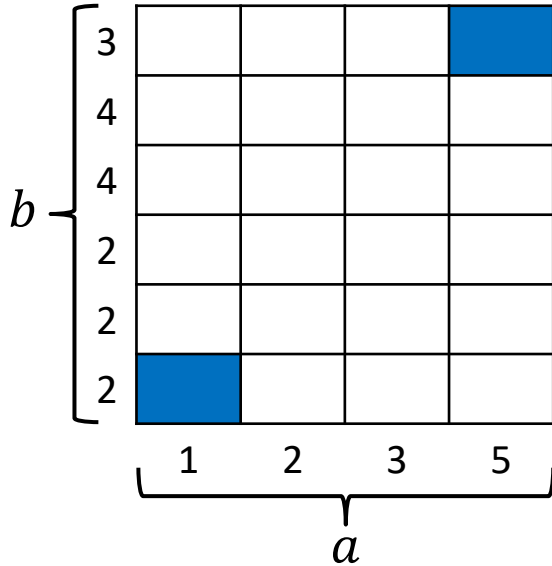
Dynamic Time Warping

Dynamic Time Warping: Rules

- **Goal:** minimize $D(a, b) = \min_{\phi} \sum_k d(a_{\phi(k)}, b_{\phi(k)})$
- **Rules** (given two sequences $\mathbf{a}$ and $\mathbf{b}$):
 - Every index of $\mathbf{a}$ must be matched with one or more indices from $\mathbf{b}$, and vice versa
 - The first index from $\mathbf{a}$ must be matched with the first index from $\mathbf{b}$ (but it does not have to be its only match)
 - The last index from $\mathbf{a}$ must be matched with the last index from $\mathbf{b}$ (but it does not have to be its only match)
 - The mapping of the indices from $\mathbf{a}$ to indices from $\mathbf{b}$ must be monotonically increasing, and vice versa, i.e. if $j > i$ are indices from $\mathbf{a}$, then there must not be two indices $m > n$ in $\mathbf{b}$, such that index i is matched with index m and index j is matched with index n , and vice versa

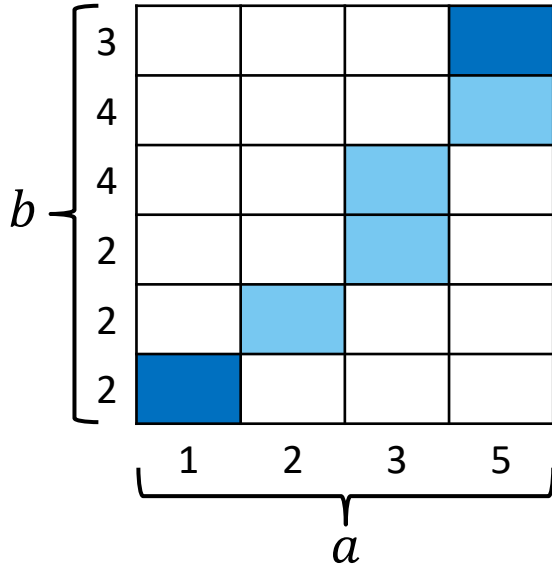
Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



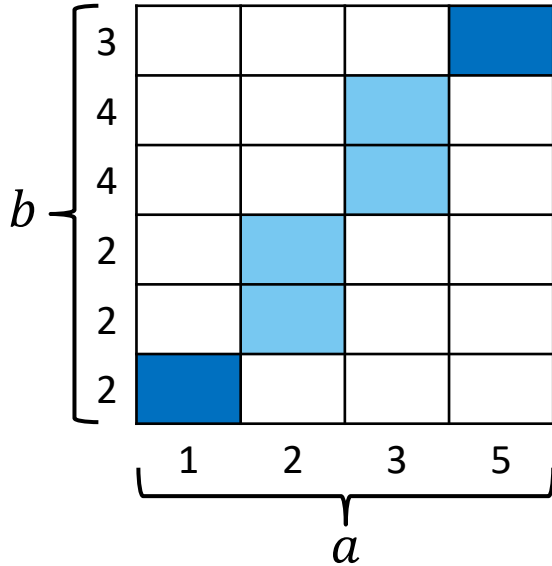
Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



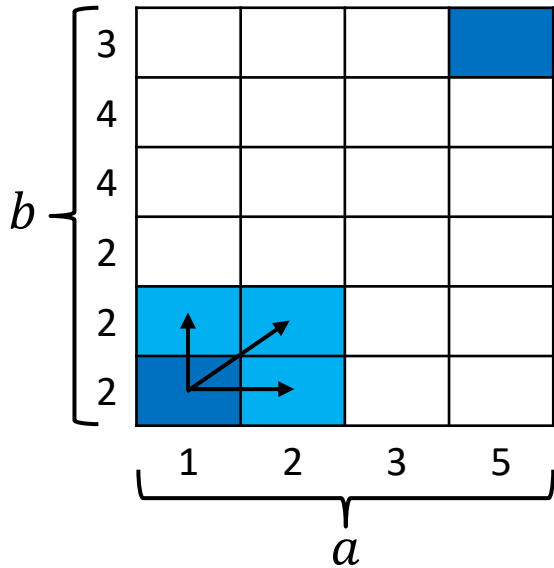
Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



Dynamic Time Warping: Possible Paths

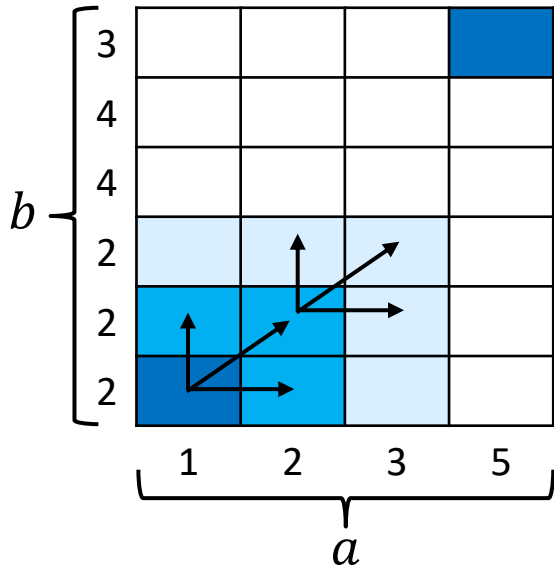
$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



- Three possible paths from each square

Dynamic Time Warping: Possible Paths

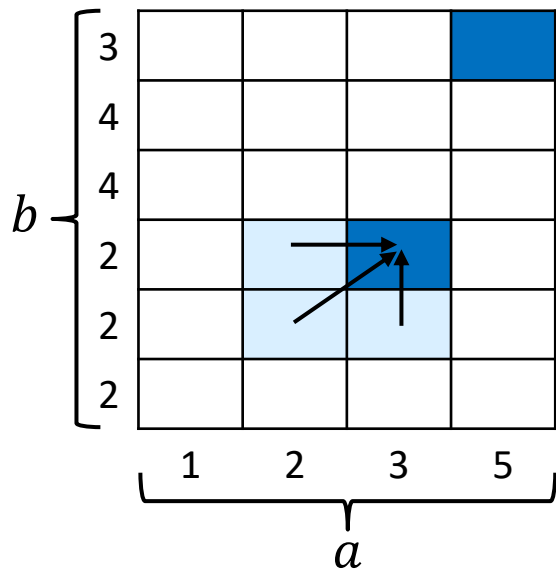
$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



- Three possible paths from each square
 - Every choice leads to three more possible paths
- ➡ $\approx 3^{4 \cdot 6}$ options

Dynamic Time Warping: Minimum Path

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



- For any cell C (matching indices i, j): three possible precursor cells
- Minimum cost (distance) for getting to C

$$d(i, j) + \min(D(i-1, j), D(i-1, j-1), D(i, j-1))$$

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3				
	4				
	4				
	2				
	2				
	2				
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3				
	4				
	4				
	2				
	2				
	2	1			
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2			
	2	1			
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2			
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2	?		
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2	1		
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2	1	?	
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2	1	2	
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2	1	2	?
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11			
	4	9			
	4	6			
	2	3			
	2	2	1	2	5
	2	1	1	2	5
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	11	6	3	5
	4	9	5	3	3
	4	6	3	2	3
	2	3	1	2	5
	2	2	1	2	5
	2	1	1	2	5
		1	2	3	5
		a			

Goal

2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

Goal

3	11	6	3	5
4	9	5	3	3
4	6	3	2	3
2	3	1	2	5
2	2	1	2	5
2	1	1	2	5
	1	2	3	5

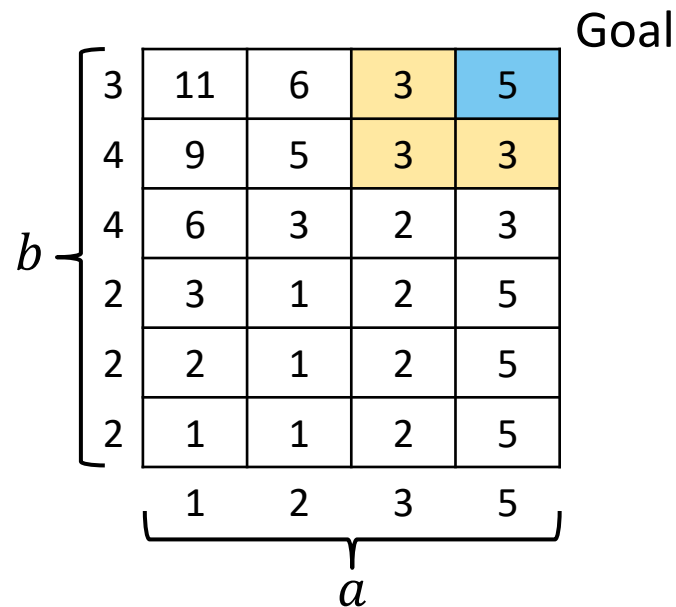
a

b

2. Compute minimum distance path

Dynamic Time Warping: Example

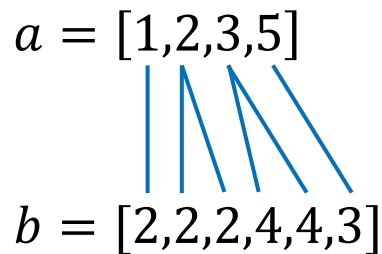
$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



2. Compute minimum distance path

Dynamic Time Warping: Example

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



Goal

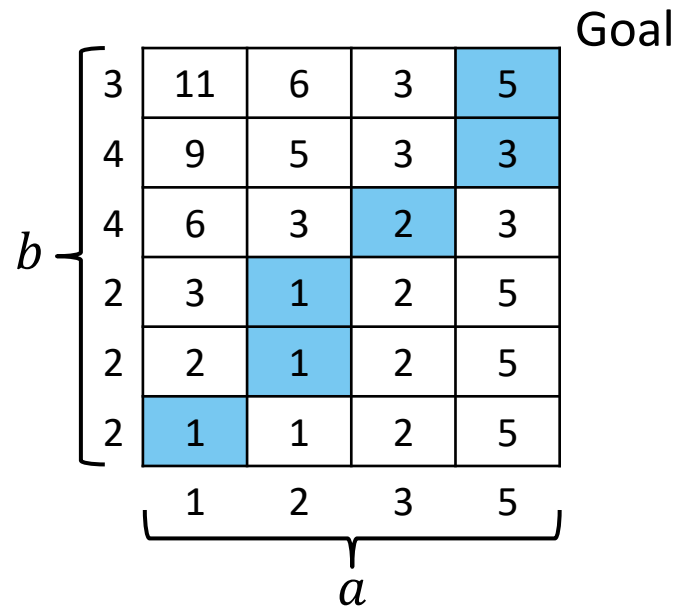
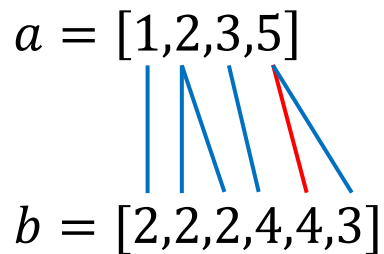
3	11	6	3	5
4	9	5	3	3
4	6	3	2	3
2	3	1	2	5
2	2	1	2	5
2	1	1	2	5
	1	2	3	5

a

2. Compute minimum distance path

Dynamic Time Warping: Example

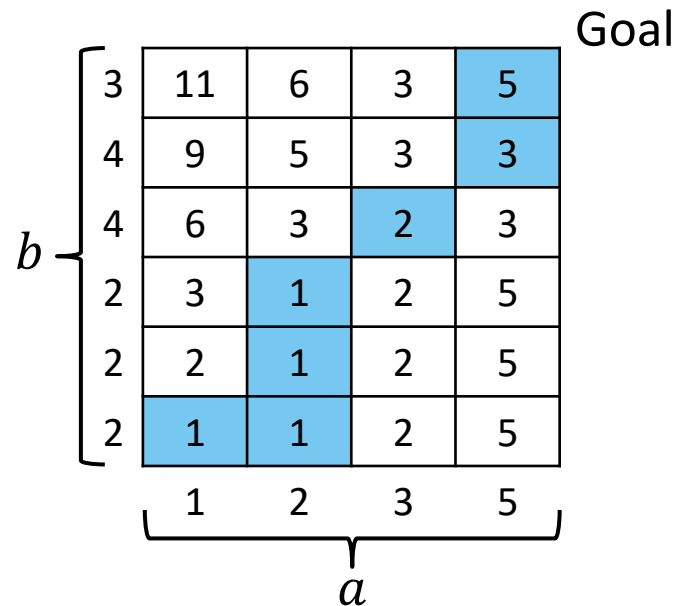
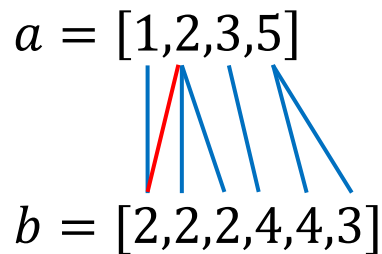
$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



2. Compute minimum distance path

Dynamic Time Warping: Example

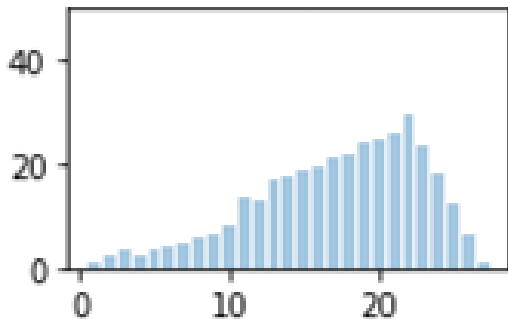
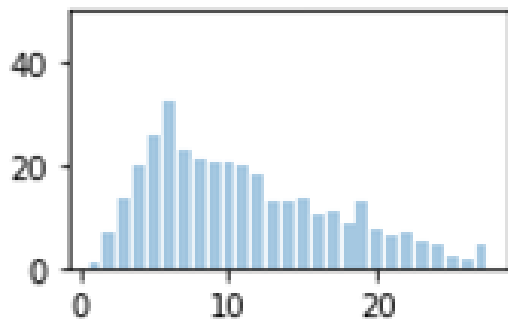
$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$



2. Compute minimum distance path

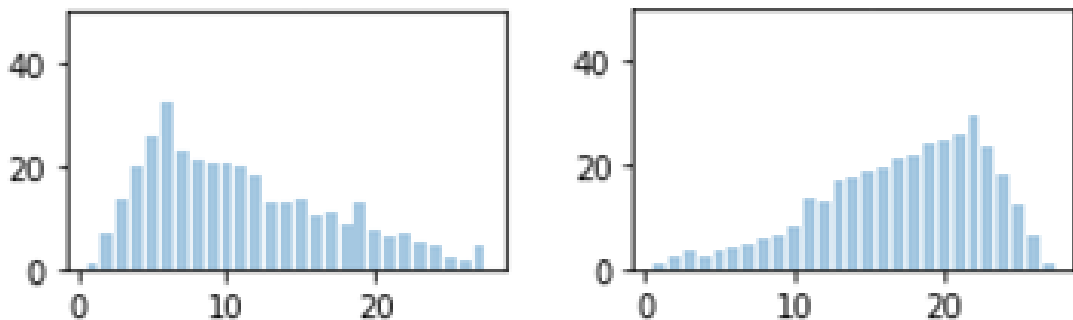
Dynamic Time Warping: Window

- Sometimes, we might want to constrain the mapping



Dynamic Time Warping: Window

- Sometimes, we might want to constrain the mapping



- We introduce a **window size** w : an element in sequence $\mathbf{a}$ at index i can only be mapped to elements at index $i - w, \dots, i + w$ in sequence $\mathbf{b}$

Dynamic Time Warping: $w = 2$

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3				
	4				
	4				
	2				
	2				
	2				
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: $w = 2$

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	-			
	4	-			
	4	-			
	2	3			
	2	2			
	2	1			
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: $w = 2$

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	-	-		
	4	-	-		
	4	-	3		
	2	-	1		
	2	2	1		
	2	1	1		
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: $w = 2$

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

b	3	-	-	-	
	4	-	-	3	
	4	-	-	2	
	2	-	1	2	
	2	2	1	2	
	2	1	1	2	
		1	2	3	5
		a			

2. Compute minimum distance path

Dynamic Time Warping: $w = 2$

$$a = [1, 2, 3, 5] \quad b = [2, 2, 2, 4, 4, 3]$$

b	3	2	1	0	2
	4	3	2	1	1
	4	3	2	1	1
	2	1	0	1	3
	2	1	0	1	3
	2	1	0	1	3
		1	2	3	5
		a			

1. Compute pairwise distances

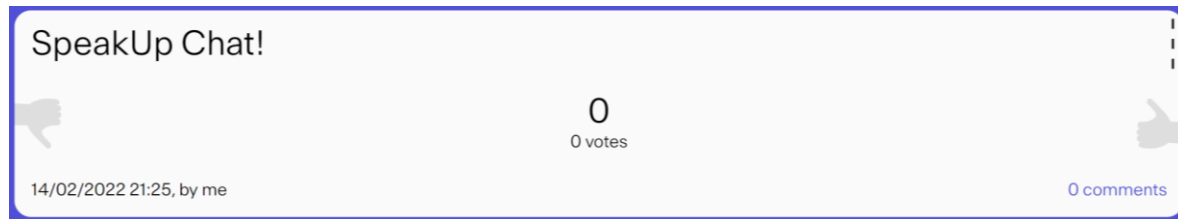
b	3	-	-	-	5
	4	-	-	-	3
	4	-	-	2	3
	2	-	1	2	5
	2	2	1	2	5
	2	1	1	2	-
		1	2	3	5
		a			

2. Compute minimum distance path

Your Turn – Dynamic Time Warping

Run spectral clustering using DTW with a window size of $w = 3$:

- How do the results differ from previous results?
- What happens if you set $w = 0$?
- And if you set $w = 27$?

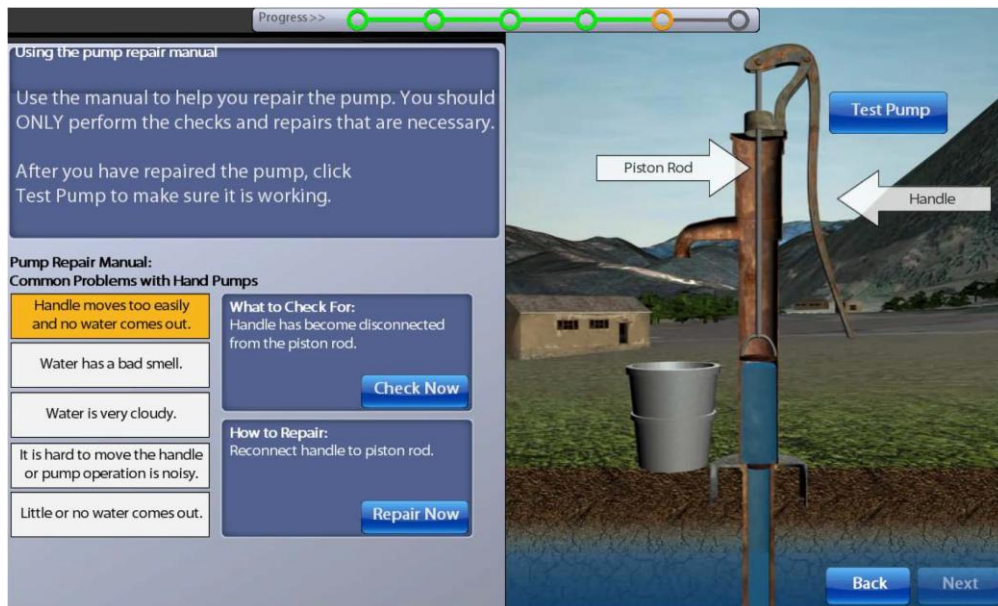


<https://go.epfl.ch/mlbd-speakup>

Agenda

- Aggregating features over time
- Defining fixed time intervals (weeks, levels in a game, etc.)
- Dynamic Time Warping
- **String Metrics**
- Markov Models

Example from Research: String Metrics



$C1 \rightarrow C2 \rightarrow C3 \rightarrow C4 \rightarrow R4 \rightarrow P \rightarrow C5 \rightarrow R5 \rightarrow P$

$C4 \rightarrow R4 \rightarrow P \rightarrow C5 \rightarrow R5 \rightarrow P$

Example from Research: String Metrics

- **Levensthein distance**: minimal number of single character edits (insertion, deletion, substitution) to change one string into the other
- **Longest common subsequence (LCS)**: string similarity measure, find the longest common subsequence between two sequences

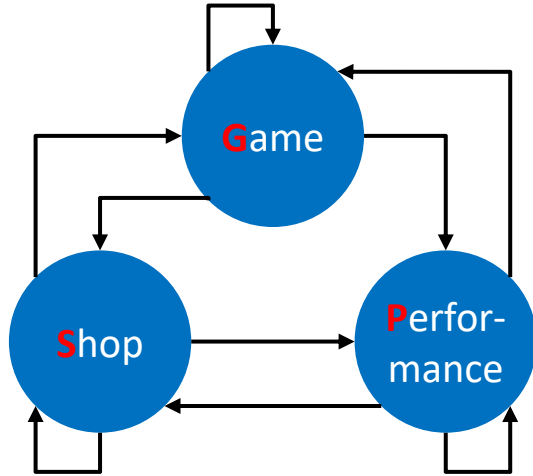
Agenda

- Aggregating features over time
- Defining fixed time intervals (weeks, levels in a game, etc.)
- Dynamic Time Warping
- String Metrics
- **Markov Models**

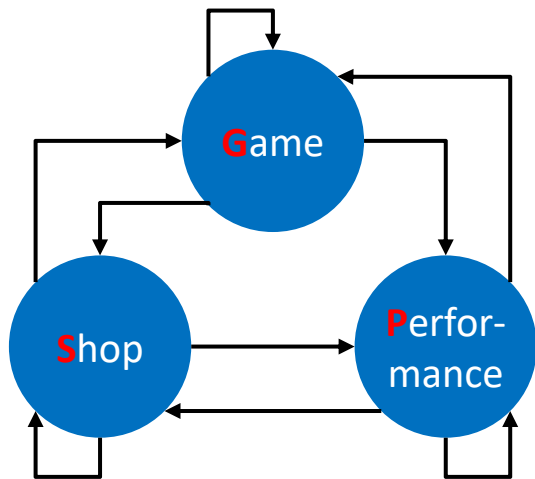
Markov Models

- Detailed action sequences provide rich temporal information
- Might contain a considerable amount of noise
- We might be interested not in the detailed sequence, but in patterns (which actions tend to follow each other)

Markov Models



Markov Models



$G \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow S \rightarrow G$

$G \rightarrow G \rightarrow G \rightarrow G \rightarrow P \rightarrow G \rightarrow G$

$G \rightarrow P \rightarrow S \rightarrow G \rightarrow P \rightarrow S$

Parameters: Maximum Likelihood Estimation

$G \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow G \rightarrow$
 $P \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow G \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow S$

$$p(S|G) = \frac{10}{15} = 0.67$$

$$p(G|G) = \frac{2}{15} = 0.13$$

$$p(P|G) = \frac{3}{15} = 0.20$$

	<i>G</i>	<i>S</i>	<i>P</i>
<i>G</i>	0.13	0.67	0.20
<i>S</i>	0.79	0.11	0
<i>P</i>	0.33	0.67	0

Stationary Distribution

$G \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow G \rightarrow$
 $P \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow G \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow S$

$$\begin{array}{c} \mathbf{G} \\ \mathbf{S} \\ \mathbf{P} \end{array} \begin{array}{c} \mathbf{G} \quad \mathbf{S} \quad \mathbf{P} \\ \left(\begin{array}{ccc} 0.13 & 0.67 & 0.20 \\ 0.89 & 0.11 & 0 \\ 0.33 & 0.67 & 0 \end{array} \right) \end{array}$$

$$\pi T = \pi$$

Stationary Distribution

$G \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow G \rightarrow S \rightarrow G \rightarrow P \rightarrow G \rightarrow$
 $P \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow S \rightarrow G \rightarrow G \rightarrow S \rightarrow G \rightarrow S \rightarrow S \rightarrow G \rightarrow S$

$$\begin{array}{c} \mathbf{G} \\ \mathbf{S} \\ \mathbf{P} \end{array} \begin{array}{c} \mathbf{G} \quad \mathbf{S} \quad \mathbf{P} \\ \left(\begin{array}{ccc} 0.13 & 0.67 & 0.20 \\ 0.89 & 0.11 & 0 \\ 0.33 & 0.67 & 0 \end{array} \right) \end{array}$$

$$\pi \mathbf{T} = \pi$$

$$\pi = [0.48 \quad 0.43 \quad 0.09]$$

Expected Frequencies

- When sequences get very long (n gets large), how often do we expect to observe the transitions?

$$\begin{matrix} & \mathbf{G} & \mathbf{S} & \mathbf{P} \\ \mathbf{G} & 0.06 & 0.32 & 0.10 \\ \mathbf{S} & 0.38 & 0.05 & 0 \\ \mathbf{P} & 0.03 & 0.06 & 0 \end{matrix}$$

Distance Metrics

- Based on **Frobenius Norm**: equivalent to Euclidean distance over vectors

$$D_2(A, B) = \sqrt{\sum_{i=1}^n \sum_{j=1}^m (a_{ij} - b_{ij})^2}$$

Distance Metrics

- **Kullback-Leibler Divergence**: measures difference between two probability distributions

$$D_{KL}(P||Q) = \sum_{x \in X} P(x) \cdot \log\left(\frac{P(x)}{Q(x)}\right)$$

- **Jensen-Shannon Divergence**: measures difference between two probability distributions

$$D_{JS}(P||Q) = \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(M||Q) \quad M = \frac{1}{2} (P + Q)$$

Distance Metrics

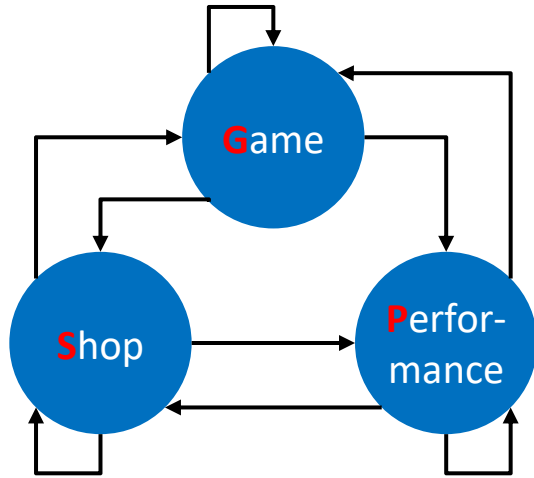
- **Hellinger Distance:** measures difference between two probability distributions

$$D_H(P||Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^n (\sqrt{p_i} - \sqrt{q_i})^2}$$

Distance between samples: Options

- Compute distance between stationary distributions: use Hellinger Distance (or Jensen-Shannon Divergence)
- Compute distance between transition matrices: use Frobenius Distance
- Compute distance between expected frequencies: use Hellinger Distance (or Jensen-Shannon Divergence)

Example from Research: Spelling Learning

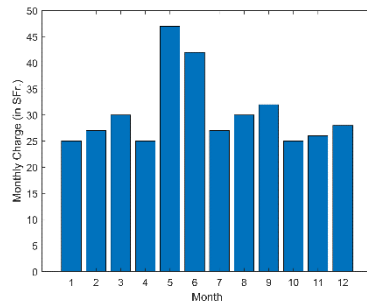


Three clusters:

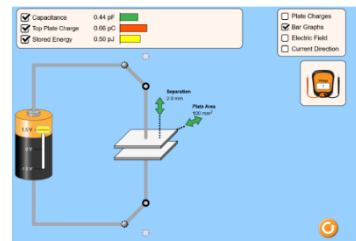
- Focused on the task
- Children, who frequently check performance/shop in-between tasks
- Spend long amounts of time off-task

Summary - Handling Time Series Data

1. Aggregating features over time
2. Defining fixed time intervals (weeks, levels in a game, etc.)
3. Dynamic Time Warping



-
4. String Measures
 5. Markov Models



Action Sequences

Administrative

The final exam will be on

Thursday 25.06.2026 from 09h15 to 12h15